

Design, Simulation, and Optimisation of a Quarter Car Suspension System

A Personal Engineering Project exploring CAD, FEA, and
Dynamic Simulation

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Nomenclature

Variables

Symbol	Definition	Units
c_s	spring damper coefficient	[Ns/m]
c_u	tyre damper coefficient	[Ns/m]
F_{damper}	damper force	[N]
F_{spring}	spring force	[N]
H	speed bump height	[m]
k_s	spring stiffness	[N/m]
k_u	tyre stiffness	[N/m]
L	speed bump length	[m]
m_s	sprung mass	[kg]
m_u	unsprung mass	[kg]
v	velocity	[m/s]
x_g	input displacement	[m]
$\dot{x}_g$	input velocity	[m/s]
x_s	vertical displacement of sprung mass	[m]
$\dot{x}_s$	vertical velocity of sprung mass	[m/s]
$\ddot{x}_s$	vertical acceleration of sprung mass	[m/s ²]
x_u	vertical displacement of unsprung mass	[m]
$\dot{x}_u$	vertical velocity of unsprung mass	[m/s]
$\ddot{x}_u$	vertical acceleration of unsprung mass	[m/s ²]
ζ	damping ratio	[-]

Acronyms

Symbol	Definition
2-DOF	2 Degrees of Freedom
CAD	Computer Aided Design
FEA	Finite Element Analysis

1. Introduction/abstract

1.1. Project objectives

A suspension system in a vehicle contains several components which work together to minimise the impact of bumps in the road. It is essential in creating a smooth ride for passengers ensuring comfort and reducing fatigue. [1] It is critical for vehicle control and stability as it keeps the tyres in contact with the road which maximises traction which is essential for steering, braking and cornering. [2] Therefore, in high performance vehicles such as formula car, solar races, GT cars etc it is vital to have this system optimised.

A quarter car model is often used in optimising these systems as it reduces complexity which allows engineers to analyse vertical motion, vibration and suspension response without having to model the entire car. [3]

The Core Objectives:

- **Develop a mathematical model** representing the masses, springs and dampers of the system.
- **Simulate the system's dynamic response** to a road disturbance such as a bump using Python.
- **Analyse performance metrics** such as settling time and peak body acceleration to determine an optimal damping ratio.
- **Calculate the peak forces** of the spring and the damper using the optimised parameters.
- **Translate the parameters to a physical CAD geometry**, designing individual components to match required structural and spatial constraints.
- **Perform FEA** on the modelled components to evaluate structural integrity of components under load and then optimise the design for weight and safety.

1.2. Scope of project

This study focuses on the vertical dynamics of a wheel assembly: the sprung and unsprung mass. Linear spring and damping characteristics are being analysed under a transient input.

The study focuses on the vertical dynamics of a wheel assembly evaluating the interactions between the sprung and unsprung masses. The linear spring and damping characteristics are analysed under a disturbance input

Assumptions:

- **Linear system properties:** The spring rate (k_s), tyre stiffness (k_u), tyre damping (c_u) are all modelled as linear constants
 - **Justification:** Physical springs and rubber tyres are naturally non-linear but they act linearly when the suspension only moves a small amount (eg 30-50mm). Keeping these values linear allowed me to create a state space model in Python and run optimisation quickly. If non-linearity was included this

would have made the code more complex without changing the initial damping trends. [4]

- **Continuous Road Contact:** The tyre is assumed to contact the road at a single point and remain in contact with the ground.
 - **Justification:** The suspension system is based on Sheffield Eco Motorsport's SEM-03 which is designed for track circuits. Because the vehicle operates at steady speeds on flat surfaces to preserve energy, track irregularities severe enough to cause wheel lift or hop will not occur during standard operation. Assuming continuous point contact is a realistic baseline for this specific vehicle and keeps the Python model mathematically straightforward. [5]
- **Fixed Disturbance Input:** The profile and parameters of the speed bump are assumed to remain constant
 - **Justification:** Real world speed bumps often are highly irregular and unpredictable. Assuming a perfectly symmetrical, mathematically smooth shape allows the road input to be written as a clean time dependent equation in the Python code. This makes it much easier to isolate and analyse how the spring and damper behave without erratic noise from an uneven road profile. [6]

2. Research

2.1. Theory

Research was conducted to better understand suspension systems and how they are mathematically modelled. This project focuses on a double wishbone suspension system with these components:

Upper and Lower Wishbones: They work together to maintain the wheel's upright position as it moves up and down over bumps ensuring consistent tyre contact with the road. [7]. The upper wishbone is connected to the top of the knuckle, while the lower wishbone is connected to the bottom of the knuckle and the chassis. The lower wishbone also has the spring- damper unit attached to it. [8]

Spring-Damper Unit: The spring damper unit is mounted to the lower wishbone and sits in between the arms of the upper wishbone. The spring absorbs and releases the energy to smooth out the bumps and uneven surfaces. They support the weight of the vehicle and maintain the ride height. [8] The damper then absorbs the shock from the spring which slows down the spring oscillations and returns the car to a level ride.

The Hub and Wheel Assembly (Unsprung Mass): This assembly includes the physical wheel, hub, and braking components. In the mathematical model, these parts are treated collectively as a single lumped unsprung mass (m_u). [8] This simplification allows the simulation to track the vertical inertia of the entire wheel assembly as it reacts directly to the road profile. [4]

The chassis (sprung mass): This is the structural frame of the vehicle which houses the powertrain and passengers. It is important to minimise the peak body acceleration of this to ensure comfort. [8]

The Tyre: The tyre serves as the primary interface between the vehicle and the track. In a quarter-car model, it is simplified as a parallel spring-damper system defined by a vertical stiffness coefficient (k_u) and a structural damping coefficient (c_u). It acts as a high-frequency filter for minor road roughness before forces reach the main suspension geometry. [3]

Definitions:

Peak Body Acceleration: The maximum vertical acceleration experienced by the vehicle chassis (sprung mass) during a road disturbance. In automotive engineering, minimizing this value is the primary way to reduce structural stress on onboard electronics and preserve passenger ride comfort. [9]

Settling Time: The time taken for an oscillation to decay to a specified boundary. Usually 2-5% of the peak value. [10]

Stiffness: Is the resistance to elastic deformation when a load is applied. Measured in newtons per metre. [11]

Damper Coefficient: Is a constant that represents the amount of resistance a system offers to motion due to damping. It determines how quickly a vibrating or oscillating system loses its energy and comes to rest. [12]

$$c = 2\zeta\sqrt{km}$$

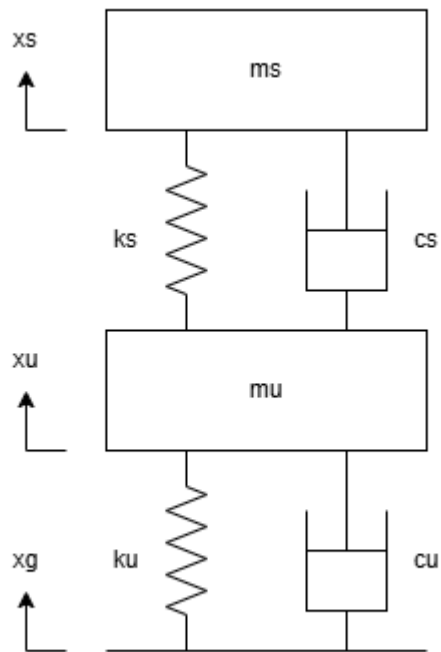
Damping Ratio: Is a dimensionless constant that quantifies how quickly an oscillation decays compared to critical damping. [13]

Sprung Mass: Parts of the car supported by the spring-damper unit e.g. chassis, driver, bodywork, battery pack, etc. [8]

Unsprung Mass: Parts not supported by the springs e.g. wheels, tyres, hubs, uprights, brake discs, part of the control arms. [8]

Two Degree-of-Freedom (2-DOF): A modeling framework that tracks two independent vertical motions simultaneously: the vertical displacement of the chassis and the vertical displacement of the wheel assembly. [14]

2.2. The Mathematical Model



Sum of Forces for the Sprung Mass:

$$\sum F_s = (\text{Eq 1})$$

$$m_s \ddot{x}_s = k_s(x_u - x_s) + c_s(\dot{x}_u - \dot{x}_s)$$

$$\ddot{x}_s = \frac{k_s}{m_s}(x_u - x_s) + \frac{c_s}{m_s}(\dot{x}_u - \dot{x}_s)$$

Sum of Forces for the Unsprung Mass:

$$\sum F_u = (\text{Eq 2})$$

$$m_u \ddot{x}_u = k_u(x_g - x_u) + c_u(\dot{x}_g - \dot{x}_u) - k_s(x_s - x_u) - c_s(\dot{x}_s - \dot{x}_u)$$

$$\ddot{x}_u = \frac{k_u}{m_u}(x_g - x_u) + \frac{c_u}{m_u}(\dot{x}_g - \dot{x}_u) + \frac{k_s}{m_u}(x_s - x_u) + \frac{c_s}{m_u}(\dot{x}_s - \dot{x}_u)$$

State Variable Substitution

Sub in x_1, x_2, x_3, x_4 :

- $x_1 = x_s$
- $x_2 = x_u$
- $x_3 = \dot{x}_s$

- $x_4 = \dot{x}_u$

Eq 1:

$$\ddot{x}_s = \frac{k_s}{m_s}(x_u - x_s) + \frac{c_s}{m_s}(\dot{x}_u - \dot{x}_s)$$

$\ddot{x}_s$ = derivative of $\dot{x}_s$ so $= \dot{x}_3$

$$\dot{x}_3 = \frac{k_s}{m_s}(x_2 - x_1) + \frac{c_s}{m_s}(x_4 - x_3)$$

Eq 2:

$$\ddot{x}_u = \dot{x}_4$$

$$\dot{x}_4 = \frac{k_u}{m_u}(x_G - x_2) + \frac{c_u}{m_u}(\dot{x}_G - \dot{x}_4) + \frac{k_s}{m_u}(x_1 - x_2) + \frac{c_s}{m_u}(x_3 - x_4)$$

Derivatives breakdown:

- $x_1 = x_s \rightarrow \dot{x}_1 = \dot{x}_3 = \dot{x}_s$
- $x_3 = \dot{x}_s \rightarrow \dot{x}_3 = \ddot{x}_s$
- $x_2 = x_u \rightarrow \dot{x}_2 = \dot{x}_4 = \dot{x}_u$
- $x_4 = \dot{x}_u \rightarrow \dot{x}_4 = \ddot{x}_u$

State-Space Representation

Put In Matrix Form:

$$\dot{X} = AX + Bu$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -\frac{k_s}{m_s} & \frac{k_s}{m_s} & -\frac{c_s}{m_s} & \frac{c_s}{m_s} \\ \frac{k_s}{m_u} & -\frac{(k_s+k_u)}{m_u} & \frac{c_s}{m_u} & -\frac{(c_s+c_u)}{m_u} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ \frac{k_u}{m_u} & \frac{c_u}{m_u} \end{bmatrix} \begin{bmatrix} x_G \\ \dot{x}_G \end{bmatrix}$$

(Note: The $u(t)$ vector points to $[x_G, \dot{x}_G]^T$)

Output Equation

$$y = Cx + Du$$

We want:

- Suspension Travel: $(x_s - x_u)$
- Chassis Velocity: $(\dot{x}_s)$

$$y = \begin{bmatrix} x_s - x_u \\ \dot{x}_s \end{bmatrix}$$

Rewrite using x_1, x_2, x_3, x_4 :

- $x_s - x_u = (1)x_1 - (1)x_2$
- $\dot{x}_s = (1)x_3$

Matrix Form:

$$\begin{bmatrix} x_s - x_u \\ \dot{x}_s \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_G \\ \dot{x}_G \end{bmatrix}$$

3. Coding

3.1. Goals

Use Python to calculate the structural boundary conditions required to size and stress test the physical components of the suspension system. To do this a set of sub goals were created:

- Set up a 2 DOF system and run the simulation on a flat road.
- Introduce a speed bump of 80mm hitting it at 10m/s at a fixed value of damping. Look for maximum displacement and peak body acceleration to determine if the suspension was exceeding its maximum mechanical travel.
- Introduce a loop to test a range of damping ratios, extracting data which meets the criteria of a settling time less than 1 second and peak body acceleration less than 0.3g.
- Use the settling time and peak body acceleration to find the optimal damping ratios
- Calculate the forces of the spring and the damper and calculate their peak values.

3.2. Simulation and Parameters

Data was collected from the Sheffield Eco Motorsports datasheet. [15]

Variable	Symbol	Value	Unit
Sprung mass	m_s	77.5	kg
Unsprung mass	m_u	10	kg
Spring stiffness	k_s	11,000	N/m
Tyre stiffness	k_u	150,000	N/m
Spring damper coefficient	c_s	1295 (used for initial testing)	Ns/m
Tyre damper coefficient	c_u	49	Ns/m

Speed bump height	H	0.08	m
Speed bump length	L	0.8	m
Vehicle velocity	v	10	m/s

A baseline spring damper coefficient of 1295 Ns/m was used initially. This assumes an approximate damping ratio of 0.7.

$$c_c = 2\sqrt{k_s m_s} = 2\sqrt{11,000 \times 77.5} \approx 1846.6 \text{ Ns/m} \quad \frac{c}{c_c} = \zeta = \frac{1295}{1846.6} \approx 0.701$$

This damping coefficient is the standard textbook starting point for testing because it offers the best balance between settling time and preventing massive overshoots. [5]

A tyre damper coefficient of 49 Ns/m was also chosen. Tyres have high stiffness but very low internal damping. Using the values the critical damping was:

$$c_c = 2\sqrt{k_u m_u} = 2\sqrt{150,000 \times 10} \approx 2449.5 \text{ Ns/m} \quad \frac{c}{c_c} = \zeta = \frac{49}{2449.5} \approx 0.02$$

A damper coefficient of 49 Ns/m sets the damping ratio to around 0.02. This low value matches empirical manufacturer data and vehicle dynamics literature, which places typical tyre damping between 1% and 5%. [5]

3.3. Original code

An original code was used to set up the 2 DOF system on a flat road to ensure it was modelled correctly. A fixed value of sprung damping coefficient was used. This value was 1295 Ns/m which was calculated by using a damping ratio of 0.7. As you can see from figure 1, the graph remains at 0 for chassis displacement, wheel displacement, suspension travel and chassis velocity. This validates the model and this section of the code.

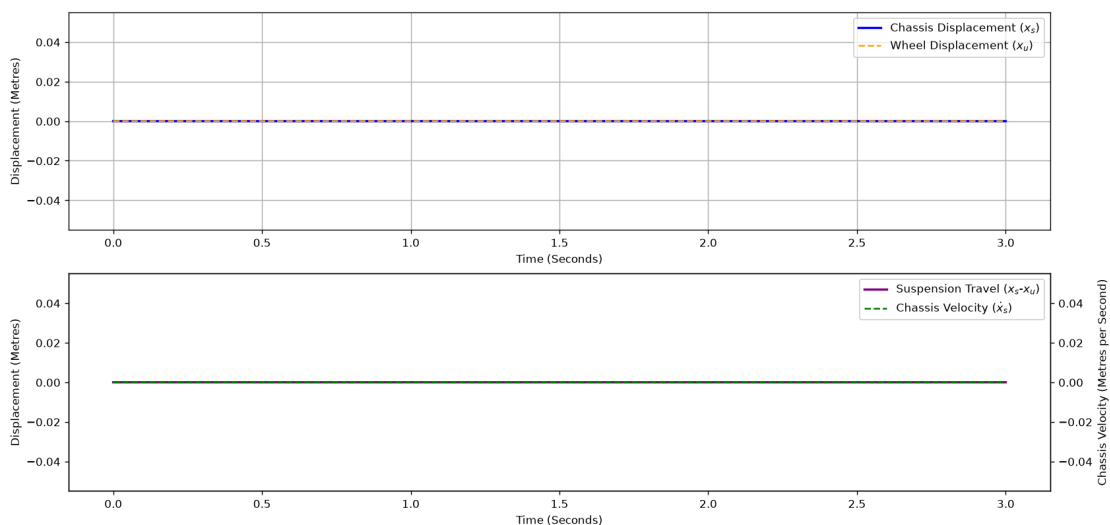


Figure 1: Dynamic Response for a Flat Road

3.4. Road Disturbance Added

A speed bump of a 1 - cosine profile was then added to the code with $H=0.08\text{m}$ $L=0.8$ and $v=10\text{m/s}$. The simulation was run at the fixed damping coefficient of 1295Ns/m . As shown in Figure 2, the dynamic transient response of the system follows the expected physical behaviors of an underdamped 2-DOF vehicle model:

Wheel Displacement: Because the wheel assembly is much lighter than the chassis, it reacts quickly to the road. It closely follows the shape of the speed bump, peaking slightly above 0.08m due to momentum, and flattens out almost immediately after clearing the bump.

Chassis Displacement: The heavier chassis has more inertia, so its movement lags behind the wheel. The suspension successfully absorbs the shock, reducing the chassis's peak displacement to around 0.05m (well below the 0.08m bump height) and preventing any continuous bouncing.

Suspension Travel: When the wheel hits the bump first, the suspension immediately compresses, which causes the initial downward spike on the graph. As the car clears the bump, the spring rebounds and expands before quickly settling back to zero.

Chassis Velocity: The vertical velocity spikes as the chassis moves up and down over the bump. The damper quickly removes this kinetic energy, bringing the velocity back to 0m well before the 1.5 second mark.

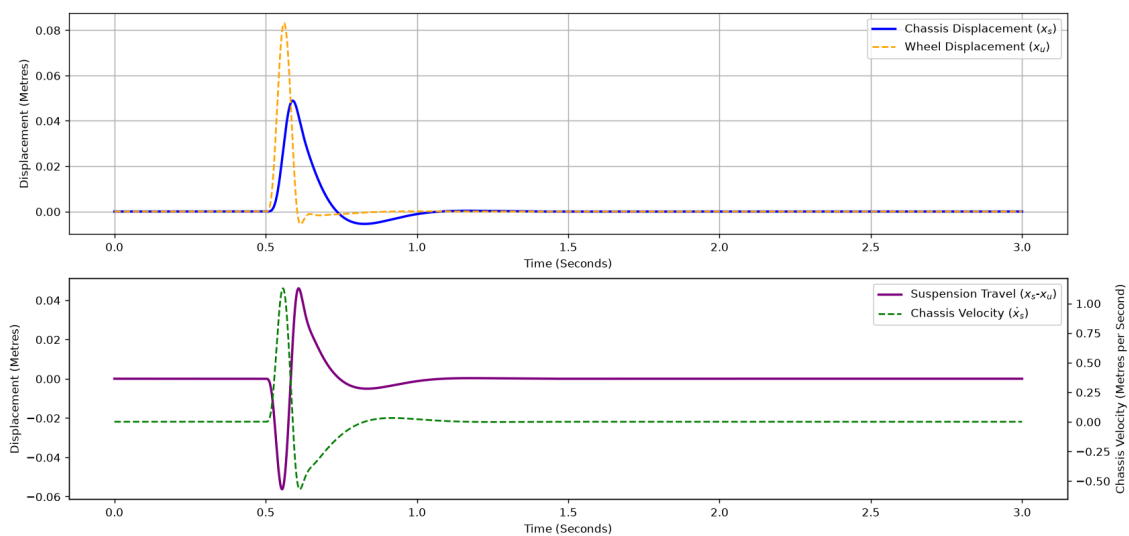


Figure 2: Dynamic Response for a Road Disturbance

3.5. Damping Ratio Iteration and Force Extraction

To identify the optimal suspension setup, the damping ratio was tested iteratively across a range of 0.05 to 1.5. The initial target was to find a configuration where the settling time was less than 1 second and peak body acceleration was less than 0.3g.

However, running the parametric sweep revealed that meeting both strict criteria at the same time is mathematically impossible with this passive setup.

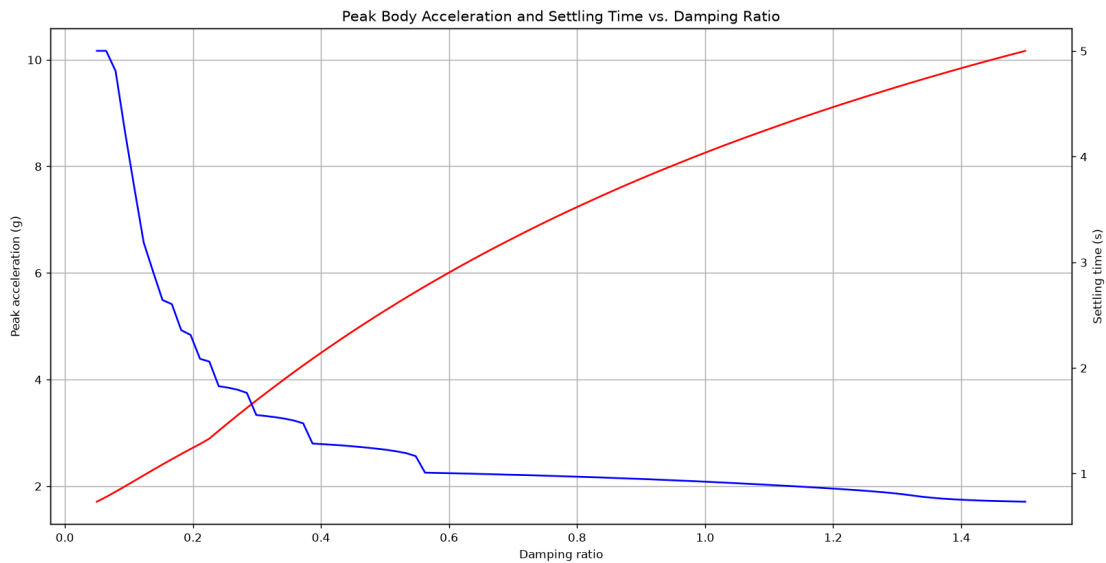


Figure 3: Peak Body acceleration and Settling Time against Damping ratio

As shown in Figure 3, suspension design requires a direct trade-off. As the damping ratio increases, the peak acceleration curve (red) climbs steeply, while the settling time curve (blue) drops down. Because these curves move in opposite directions, a compromise had to be made based on three distinct design scenarios:

- Scenario 1: Minimise peak body acceleration
- Scenario 2: Minimise settling time
- Scenario 3: Balanced approach

Scenario 1 would allow the suspension to be optimised for comfort ensuring passengers do not experience fatigue. However, this does lead to a high settling time which would severely affect the handling. Scenario 2 optimises for handling and speed as this reduces the time it takes for oscillations to settle down ensuring the tyres have maximum contact with the road. Finally Scenario 3 uses a balanced, more realistic approach to achieve the “best of both worlds”. Rather than having a perfect balance of the two, it was decided to still have the settling time within the criteria to ensure the handling was optimised as this would be

important for races. Therefore the code calculated the peak body acceleration and damping ratio based on a settling time of 0.999 seconds.

	Scenario 1	Scenario 2	Scenario 3
optimal zeta	0.05	1.5	0.636
settling time	5	0.734	0.999
peak acceleration	1.706	10.165	6.244
max spring force	1124.799	488.346	724.339
max damper force	644.517	7757.507	4824.577
Max shock force	1296.694	7728.128	4747.349

The simulation highlights a clear shift in how structural forces develop within the assembly depending on the damping setup. In scenario 1, the spring force dominates because the lack of damping resistance allows the suspension to compress heavily. In scenarios 2 and 3, the damper forces dominate because the shock absorber generates force proportional to velocity.

Crucially the maximum spring force and maximum damper force do not occur at the same time. The peak spring force occurs at maximum physical compression, whereas the peak damper force occurs much earlier when the vertical velocity is highest.

Therefore, adding the two maximum values would create unrealistically high, inaccurate boundary conditions. To fix this, the code calculated the true combined force vector ($F_{\text{spring}} + F_{\text{damper}}$) at every single step. The resulting maximum combined shock forces represent the true peak value which will be used to test the CAD parts in FEA.

3.6. Coding Conclusions

To validate the Python simulation, the model was translated into MATLAB using an identical setup. The MATLAB script yielded identical results across all states, successfully validating the state space formulation.

Because the initial target of a peak body acceleration less than 0.3g and a settling time less than a second could not be achieved simultaneously, three compromise scenarios were established. All three scenarios will be imported into FEA to evaluate how optimising for different performance parameters affects the structural integrity of the suspension assembly. Finally, scenario 3, the balanced approach, will be used for the topology optimisation phase.

4. CAD

4.1. CAD goals

The main purpose of the CAD phase is to turn the numbers from the 2-DOF simulation into a realistic, working double-wishbone suspension system in Fusion 360. Before any parts can

be stress tested or optimised, the CAD assembly must achieve a few specific engineering goals:

Match the Specifications: The assembly layout must strictly follow the physical constraints given in the project datasheet, including the track width, ride height, wheel rim packaging, and chassis mounting spaces.

Allow Proper Movement: Every part must be jointed correctly so the suspension can move smoothly up and down through its full mechanical travel without the components colliding or binding up.

Set Up for FEA Stress Testing: The structural components, specifically the wishbone assemblies need clean, accurate geometry. This ensures they can be cleanly imported into ANSYS to be tested.

Design with Optimisation in Mind: The baseline parts need a solid, realistic shape that gives ANSYS Topology Optimisation enough material to work with later on.

4.2.Chassis and Skeleton

To aid in transitioning the simulation to a physical CAD model, a hardpoint skeleton was created from measuring points from available models. The origin was taken as the wheel centre (Figure 4). Using the Sheffield Eco motorsports datasheet, many values such as the track width (1500mm), wheelbase (2300mm), the tyre radius (277.4mm) and static ride height (80mm) which were then plotted onto the skeleton. The track width and wheelbase were divided by 2 as this is for a quarter car model.

From measuring distances on technical drawings, the location of each spindle was then found (44mm 389mm above the wheel centre). The distance from the centre of the tyre to the end of the knuckle was then added (135.5mm). This is to be the start of the wishbones. It was also determined from a 3D model of the chassis, there was 305mm of space to fit the spindle and chassis mount. The wishbones were decided to be at an angle of 30 degrees to provide a structurally stable base for resolving longitudinal braking loads into the chassis, while ensuring a dynamic clearance envelope that prevents the tire from rubbing against the control arms under a full 30 degree steering lock.

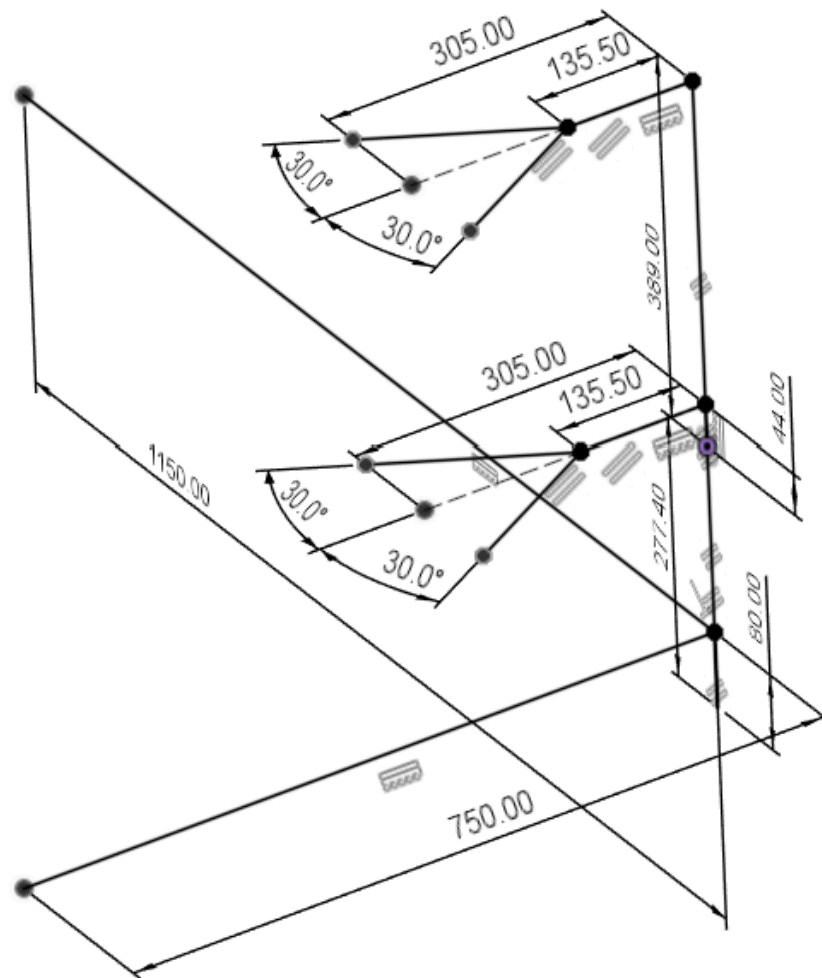


Figure 4: Suspension System Skeleton

4.3. Wheel Assembly and Knuckle

Wheel and Hub Components: The tyre, wheel rim, and wheel hub were sourced directly as reference CAD models from the team's interview task materials. Sourcing these parts directly provided a reliable physical boundary to package the rest of the suspension assembly against.

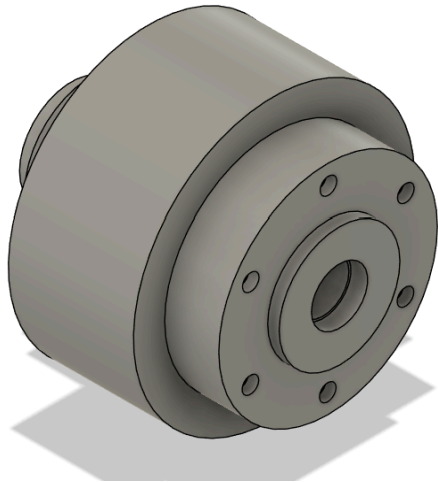


Figure 5: Wheel Hub



Figure 6: Wheel and Tyre Assembly

The Spindle: The spindle shaft which connects the hub to the upright and allows the wheel to rotate smoothly was designed from scratch in Fusion 360 using precise dimensions taken from official technical drawings.

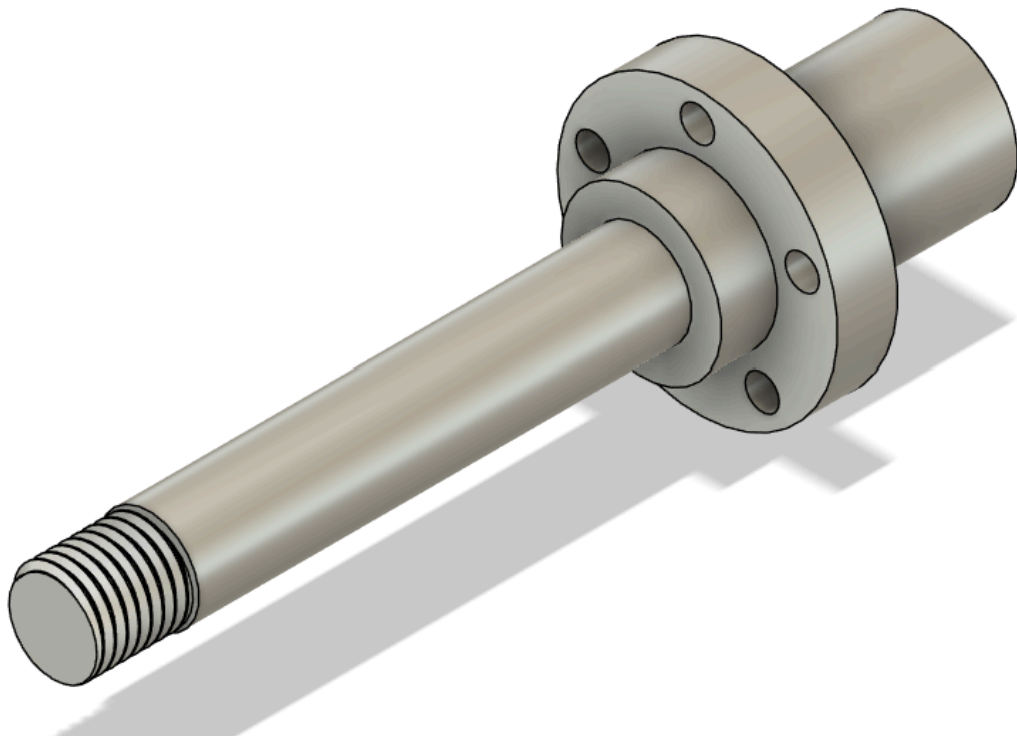


Figure 7: The spindle

The Knuckle: The upright connects the upper and lower wishbones to the spindle. This part was reverse engineered from scratch using reference images and technical drawings. The hollow box shape, the internal support ribs, and the bolt hole patterns were modelled to match the real part as closely as possible.

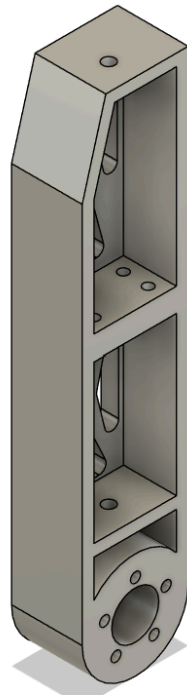


Figure 8: The Knuckle

4.4. Upper and lower wishbones

To translate the 2-DOF dynamic hardpoints into physical components, custom upper and lower wishbone assemblies were designed. A modular, multi-piece design was chosen over a single cast or bent piece. This allowed for precise manufacturing and adjustable end-connections.

The core design utilizes a three-part construction: a central mounting clevis block flanked by structural steel tubing.

The clevises serve as the structural hubs of the wishbones. The lower wishbone clevis (Figure 9) and upper wishbone clevis (Figure 10) were designed to interface directly with the knuckle upright.

- **Threading and Hardware:** To secure the spherical rod ends (heim joints) that anchor the assembly, the receiving ends of the clevis blocks are modeled with internal M10x20mm threads.
- **Design Differentiation:** While both blocks provide solid mounting paths, the lower wishbone clevis is physically beefier and features dedicated mounting bosses. This structural difference is necessary because the lower arm must directly support the massive vertical loads of the spring-damper unit, whereas the upper arm primarily manages lateral cornering forces and maintains camber control.

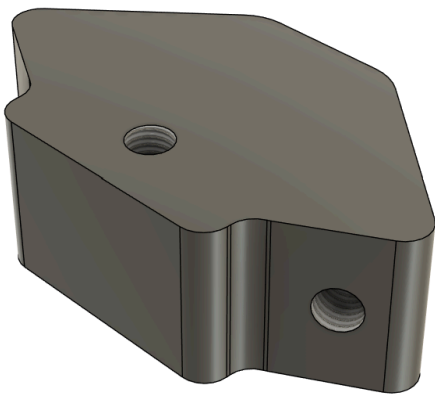


Figure 9: Lower wishbone Clevis

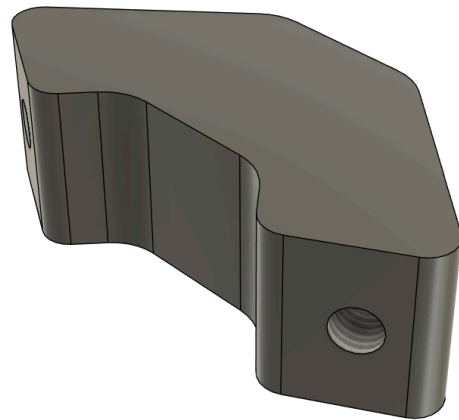


Figure 10: Upper Wishbone Clevis

Linkage Tubing and Geometry Packaging

To complete the wishbone structures, high-tensile steel tubing was integrated into the clevis blocks to form the classic "A-arm" geometry.

- **Tubing Profile:** The structural arms consist of 16mm outer diameter tubing with a 1.5mm wall thickness (16x1.5mm). This profile provides an optimal strength-to-weight ratio against bending and buckling under suspension compression.
- **Packaging and Tire Clearance:** To prevent the tire from impacting the suspension arms at full steering lock, the wishbone legs must clear the swept arc of the wheel. For the 90mm wide track tire utilized in this assembly, a safe, non-binding steering envelope was achieved by angling each leg outward at exactly 30 degrees relative to the longitudinal centerline. This creates a 60 degree total included angle between the two structural tubes, providing the necessary clearance for the wheel envelope while establishing the final spacing at the chassis mounts.
- **Chassis Alignment:** Initially, the upper wishbone was designed with a downward angle toward the chassis to optimise roll-center migration. However, to simplify fabrication and ensure clean, non-binding travel during physical motion tests, the geometry was updated to a flat, parallel configuration.

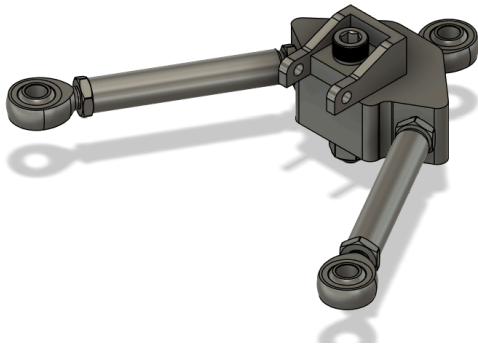


Figure 11: Lower Wishbone Assembly

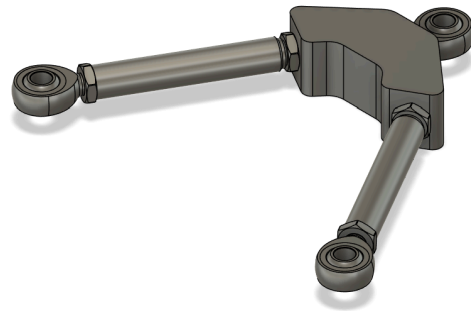


Figure 12: Upper Wishbone Assembly

Chassis Mounts (Figure 13): The wishbones connect to the chassis skeleton via robust, double-shear mounting brackets housing the spherical heim joints. This double-shear arrangement prevents bending on the mounting bolts, distributing the massive impact forces symmetrically into the chassis rails.

Lower Spring-Damper Mount (Figure 14): A specialized clevis bracket was designed and integrated onto the lower wishbone arm. This bracket dictates the physical mounting angle of the shock absorber (acting at 50 degrees relative to the lower wishbone plane), localising the motion ratio and transferring the ground forces directly from the wheel assembly into the spring-damper unit.

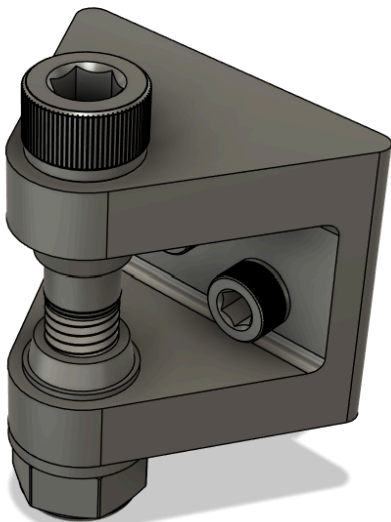


Figure 13: Chassis Mounts

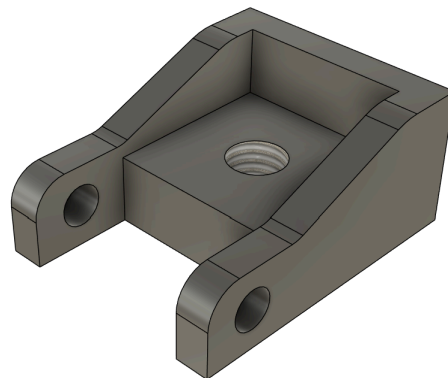


Figure 14: Lower Wishbone Spring-Damper Mount

4.5. Spring-Damper Unit

To control the dynamic travel of the suspension, a custom spring-damper unit was integrated into the assembly. This section details the complete assembly, the structural mounting interface, the governing spring math, and the physical reconstruction of the internal damper components.

The complete spring-damper assembly (Figure 15) acts as the primary energy-dissipating member of the suspension. It bridges the lower wishbone directly to the upper chassis framework.

Top Chassis Mount (Figure 16): The upper eyelet of the damper connects to a dedicated chassis mount. This mount uses a single-bolt shear pin arrangement designed to handle the direct vertical compression forces transmitted from the wheel. The pivoting joint allows the damper to rotate slightly as the wishbones travel through their motion arc, preventing bending moments on the damper shaft.

Spring Sizing Calculations: Rather than arbitrarily guessing a spring rate, the target stiffness was strictly defined as 11,000N/m, sourced directly from the Sheffield Eco Motorsport datasheet.

Material and Geometry Assumptions: To translate this stiffness into physical dimensions, the spring was assumed to be made of standard spring steel with a shear modulus (G) of 79.3GPa. Two physical packaging constraints from the damper unit defined the envelope:

- An outer diameter (OD) of 42mm (measured from the damper unit's clearance limits).
- A physical spring height of 36mm to perfectly bridge the mounting gap on the damper body.

Calculating the Number of Coils: With a chosen wire diameter (d) of 4mm, the mean coil diameter (D) was established as:

$$D = OD - d = 42 - 4 = 38mm$$

Using the standard elastic deflection equation, the required number of active coils (N) was calculated to match the datasheet stiffness of 11,000N/m / 11N/mm:

$$N = \frac{Gd^4}{8D^3k_s} = \frac{79,300 \times 4^4}{8 \times 38^3 \times 11} \simeq 4.2$$

CAD Modeling in Fusion 360: With these dimensions calculated, the physical component was modeled in CAD. The Coil feature in Fusion 360 was utilized, inputting the 4mm wire diameter, the calculated 38mm pitch diameter, the 40mm overall height, and the calculated coil count to generate a highly accurate, manufacture-ready spring geometry.



Figure 15: Spring-Damper Assembly

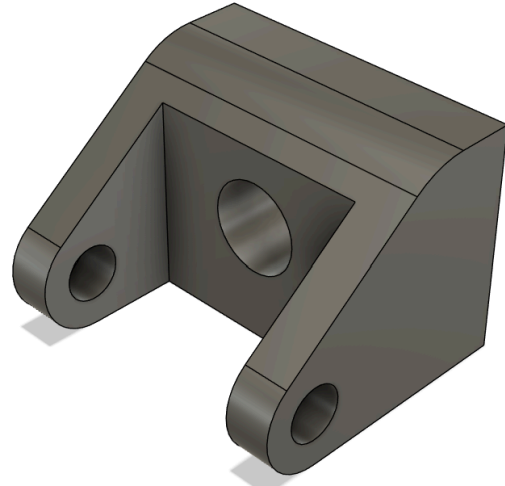


Figure 16: Upper Spring-Damper Mount

The internal damper unit (Figure 17) was modeled to accurately represent a real-world twin-tube shock absorber. Because exact technical manufacturing drawings for the damper internals were unavailable, the CAD geometry was reverse-engineered and scaled using detailed reference images of the SEM-03 to ensure correct proportions and realistic packaging.

- **Concentric Tube Design:** The unit consists of two main telescoping bodies:
 - **The Outer Tube (Damper Body):** Acts as the main reservoir for the damping fluid and provides the seating platform for the lower spring collar.
 - **The Inner Tube (Damper Shaft/Piston Rod):** Telescopes smoothly inside the outer body. This shaft must maintain a high surface finish to prevent seal wear, and its outer diameter is carefully matched to the inner clearance of the spring coils to prevent physical rubbing or interference during compression.
- **Mounting Pin Hole (Figure 18):** At the base of the outer tube, a dedicated mounting eyelet featuring a precision-bore pin hole was modeled. This pin hole secures the bottom of the shock to the bracket on the lower wishbone clevis (established in Section 4.4). By aligning this pin hole with a high-tensile bolt, the high-impulse ground forces from the upright are directly and safely transferred into the damper unit.



Figure 17: Inner Spring-Damper Tube



Figure 18: Outer Spring-Damper Tube

4.6. Assembly and motion test

With all individual sub-components modeled, including the wheel hub assembly, upper and lower wishbones, and the concentric spring-damper unit, the final stage of the CAD process was to integrate them into a unified suspension corner. This section details the mating of these components, the exploded assembly stack, and the physical motion testing used to validate the system's kinematics.

Joint Constraints and Assembly Feasibility

To replicate physical mechanics, the model was integrated using functional joints in Fusion 360 rather than rigid locks:

- **Revolute joints** at the chassis brackets allow free suspension travel.
- **Spherical joints** at the knuckle-to-clevis connections mimic the multi-axis movement of real-world heim joints.
- **A cylindrical joint** on the damper allows realistic telescoping while keeping the spring concentric.

Motion Testing and Collision Detection

A dynamic motion test was executed by cycling the assembly from full compression (bump) to full extension (droop). Using active collision detection, we verified that:

1. The spring-damper unit completely clears the upper and lower wishbones throughout its 50 degree travel stroke.
2. The 90mm wide track tire does not bind or clash against the 30 degree angled wishbone legs at full steering lock.

This motion test successfully validated both the physical packaging and the kinematic design of the suspension corner.



Figure 19: Full Suspension Assembly

4.7. CAD Conclusions

Following the finalisation of the individual component models, the complete suspension assembly has been fully mated and validated for physical clearances. This digital assembly is now prepared for direct import into ANSYS Mechanical, where it will undergo FEA and structural optimisation to minimise mass while ensuring structural integrity under design loads.

5. FEA

5.1. FEA goals

The main purpose of the Finite Element Analysis (FEA) phase is to stress-test the CAD assembly in ANSYS Mechanical to make sure the parts can handle real-world loads without breaking, and then optimize them to save weight. Specifically, the FEA phase aims to achieve the following goals:

Verify Structural Integrity: Apply the worst-case peak shock forces calculated during the 2-DOF speed bump simulation to the suspension assembly. The goal is to identify high-stress concentrations and ensure the components do not yield under impact.

Meet Target Factor of Safety (FoS): Evaluate the Equivalent (Von-Mises) Stress and Total Deformation across the upright and wishbones to ensure the entire design satisfies the team's target factor of safety.

Ensure Manufacturability: Take the optimized, organic shapes generated by ANSYS and translate them back into clean, realistic designs in Fusion 360 that can actually be manufactured using methods like CNC machining or 3D printing.

5.2. Boundary conditions

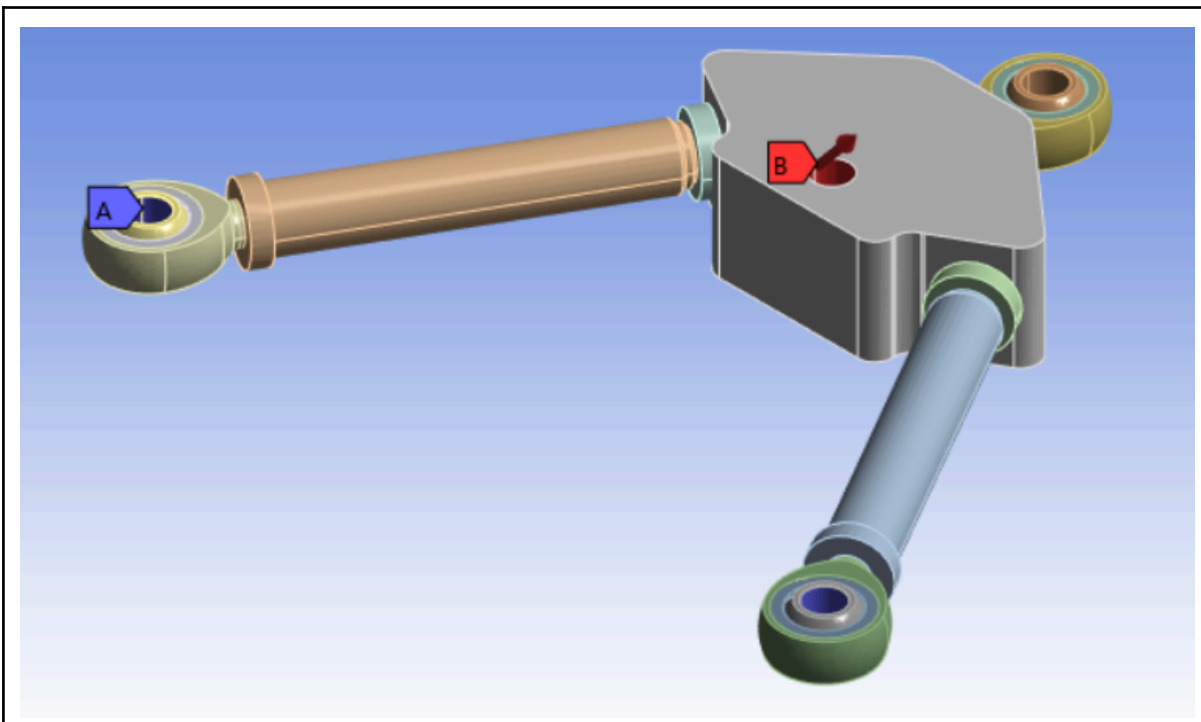


Figure 20: Simplified Lower Wishbone Assembly

To avoid convergence issues and reduce solving errors caused by complex contact definitions, the wishbone assembly was simplified. The standard hardware and loose bolt configurations were replaced with solid cylindrical pins. This ensures a clean, reliable force transfer between the separate components while keeping the mesh and contacts stable.

Supports and Constraints

The assembly is fixed to the vehicle chassis using spherical rod ends (heim joints). To recreate this constraint accurately in the FEA software:

- **Fixed Supports** were applied directly to the internal faces of both spherical rod end bearings. This anchors the wishbones to the chassis frame and prevents rigid body motion while under load.

Force Resolution and Scenarios

Because the physical spring-damper unit acts at a 50 degree angle relative to the lower wishbone plane, the total peak shock force from the 2-DOF code had to be resolved into horizontal (F_x) and vertical (F_y) vector components using basic trigonometry:

$$F_x = F_{shock} \times \cos(50)$$

$$F_y = F_{shock} \times \sin(50)$$

The forces were applied directly to the center of the mounting hole at the top of the integrated shock clevis bracket. To evaluate the entire design envelope, three separate loading scenarios tracking different compromise damping parameters were calculated and tested:

Scenario 1:

$$F_x = 1296.694 \times \cos(50) = 833.498836755$$

$$F_y = 1296.694 \times \sin(50) = 993.325233126$$

Scenario 2:

$$F_x = 7728.128 \times \cos(50) = 4967.54492447$$

$$F_y = 7728.128 \times \sin(50) = 5920.08951011$$

Scenario 3:

$$F_x = 4747.349 \times \cos(50) = 3051.53711606$$

$$F_y = 4747.349 \times \sin(50) = 3636.68032100$$

5.3. Mesh

To prepare the assembly for structural analysis, a finite element mesh was generated across the components. The mesh settings were balanced to ensure high geometric accuracy in critical zones while keeping computation times reasonable.

Global Mesh Strategy: A standard tetrahedral mesh was applied globally to the main structural linkages and the clevis block. Tetrahedral elements provide excellent flexibility for mapping the curved transitions and angled faces of the custom wishbone geometry.

Local Mesh Sizing: The spherical rod ends act as the primary load paths where the assembly anchors to the chassis. Because these areas experience high stress concentrations from the fixed constraints, a local Face Sizing control was applied directly to the rod end geometries.

Refinement Metrics: The element size at these spherical interfaces was restricted to a fine 2mm limit. As shown in Figure 21, this local sizing creates a noticeably dense mesh grid

around the pivot points, which is crucial for accurately capturing the steep stress gradients and preventing artificial stress hotspots during the solve phase.

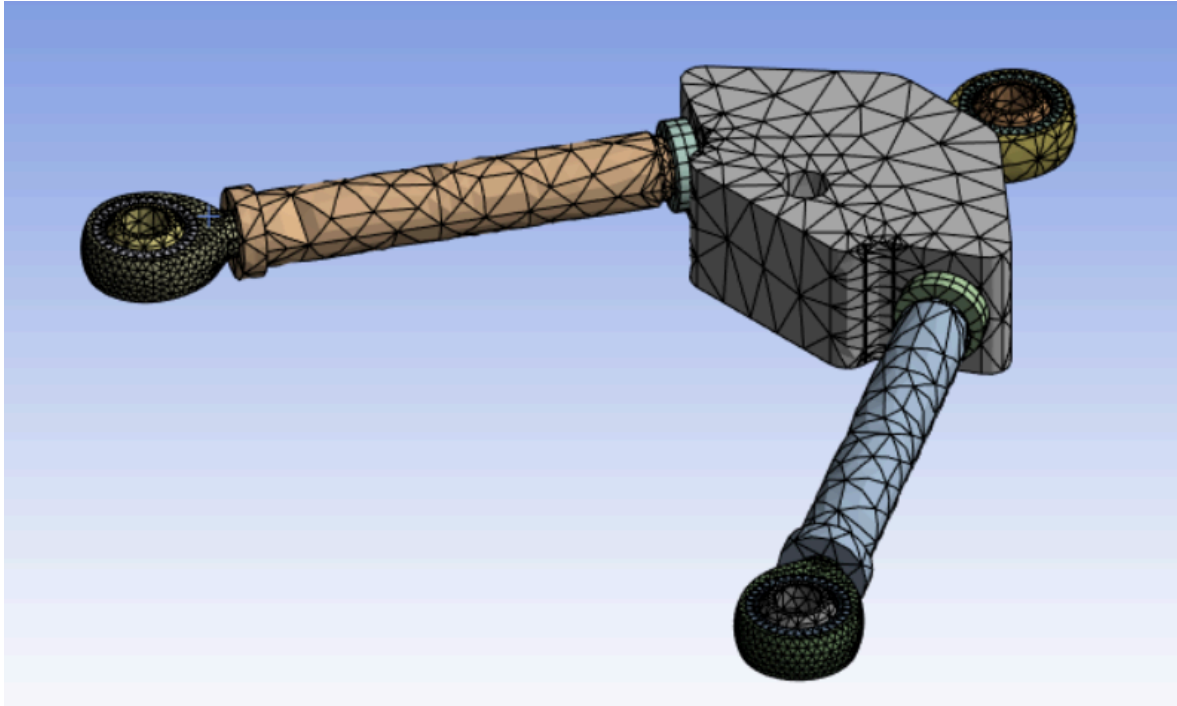


Figure 21: Mesh for Lower Wishbone

5.4. Solve

The finite element model was solved across three critical dynamic loading scenarios derived from the 2-DOF vehicle simulation. Initial evaluations focused on the baseline design, which utilized M10 spherical rod ends (heim joints) at the chassis interface.

Initial Design Evaluation & Failure Analysis

The first iteration failed to meet structural requirements, demonstrating critical stress concentrations that would result in mechanical yielding. As detailed in Table 3, Scenario 2 and Scenario 3 loading conditions induced catastrophic equivalent (Von-Mises) stress peaks of 2187.8 MPa and 1344.0 MPa, respectively far exceeding the yield strength of standard structural steels.

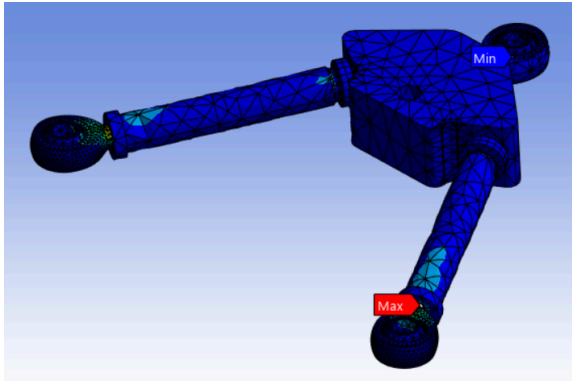


Figure 22: Stress Contours

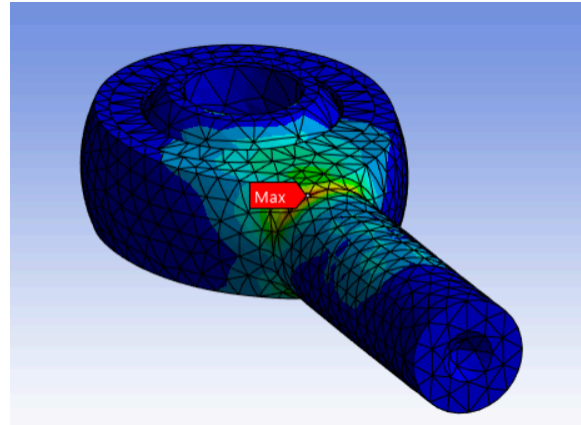


Figure 23: Zoomed in Stress Contours

- **Cantilevered Support Effect:** This structural failure was primarily driven by a geometric mismatch. The stark difference in diameter between the M10 spherical rod shaft and the main wishbone leg created an abrupt cross-sectional transition. This acted as a cantilevered support condition, forcing the unbraced thread runout area to absorb massive bending moments.
- **Nut Bracing Effects:** The initial simplification of omitting the locking nuts during contact modeling removed critical pre-load and structural clamping support. Without a nut to brace and distribute face loads against the wishbone leg, the rod end acted as a pure cantilever under impact forces.

Redesign and Second Iteration Validation

To remedy these stress concentrations, the assembly underwent a structural redesign:

1. **Thread Scale-up:** The chassis interface joints were upscaled from **M10 to M14 spherical rods** to increase the cross-sectional area and polar moment of inertia.
2. **Leg Thickening:** The wishbone receiver legs were thickened to match the M14 hardware profiles, eliminating the drastic geometric step down.

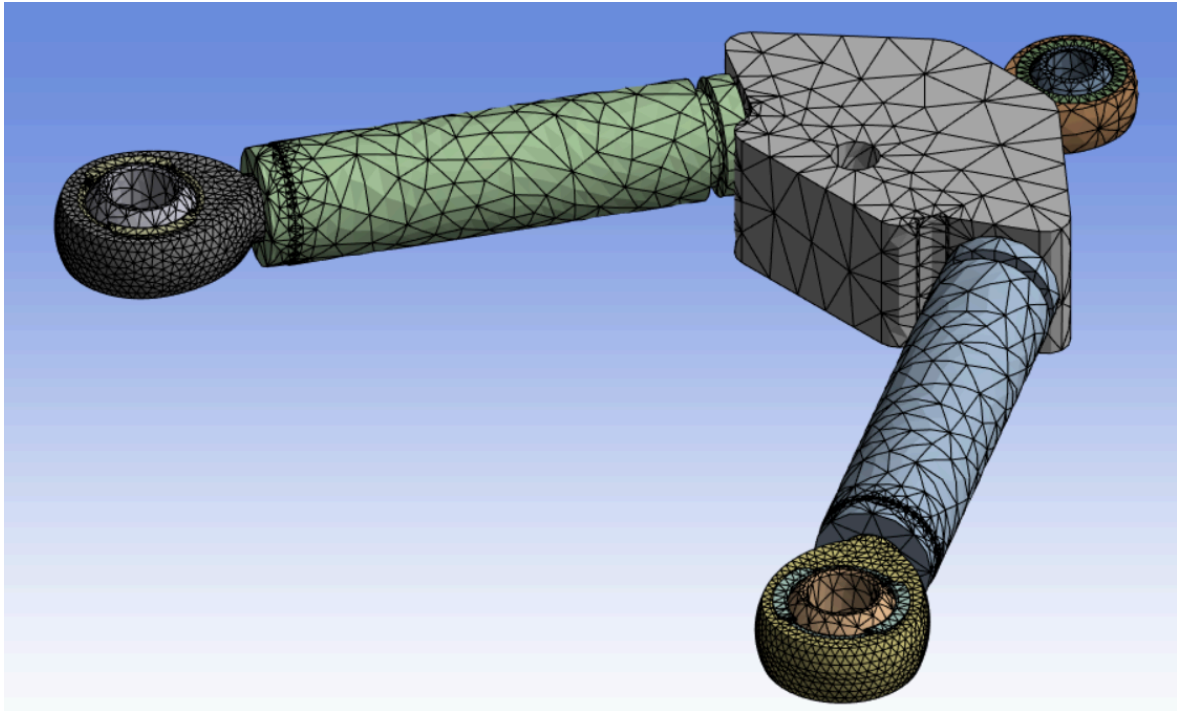


Figure 24: Mesh for Updated Lower Wishbone

3. **Contact Re-modeling:** The hardware constraints were re-integrated into the assembly, treating the integrated wishbone legs as reinforced structural components to simulate the face-bracing effect of a locked nut.

The updated design was re-solved under identical boundary conditions. Stresses dropped substantially across all scenarios, with the balanced approach (Scenario 3) yielding a manageable peak stress of 495.26 MPa.

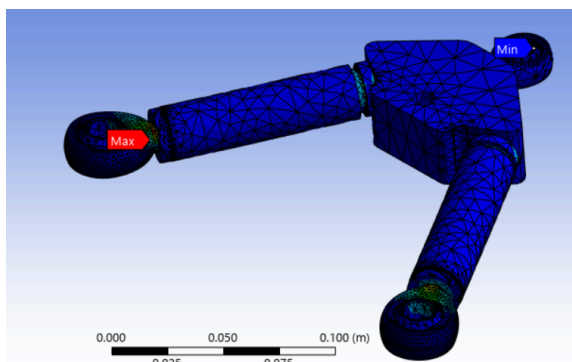


Figure 25: Stress Contours for Updated Lower Wishbone

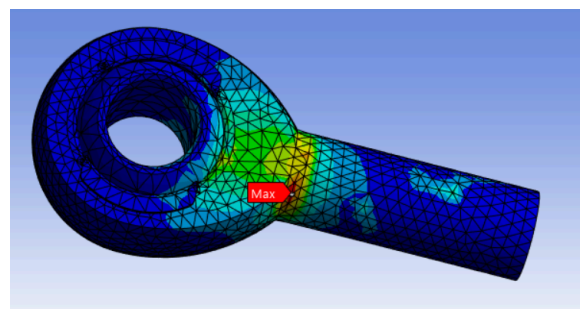


Figure 26: Zoomed in Stress Contours for Updated

Performance Metric	Scenario 1	Scenario 2	Scenario 3
Initial Design Peak Stress	367.09 MPa	2187.80 MPa	1344.00 MPa
Initial Design Max Displacement	0.499 mm	2.973 mm	1.826 mm
Updated Design (M14) Peak Stress	135.27 MPa	806.22 MPa	495.26 MPa
Updated Design (M14) Max Displacement	0.199 mm	1.186 mm	0.729 mm

5.5. Optimisation

With structural integrity verified on the robust M14 baseline design, topology optimization was conducted in ANSYS to minimize parasitic mass while maintaining system stiffness. Scenario 3 (the balanced dynamic response) was selected as the optimisation boundary envelope.

Optimization Strategy and Constraints

- **Mesh Tuning:** While a 2mm element size was targeted for optimal boundary definition, it exceeded software node limits for the optimisation solver. The global mesh size for the wishbone linkages was adjusted to 3mm to maintain compliance without compromising localised accuracy.
- **Exclusion Zones:** To ensure the component remained functional within the suspension assembly, manufacturing and assembly keep-out zones were defined. Preserved regions were locked around the four vital connecting bolt holes and the spring-damper mounting interfaces.
- **Objective Function:** The optimisation loop targeted maximum global stiffness (minimising compliance) while enforcing a strict 50% volume reduction on the design space.

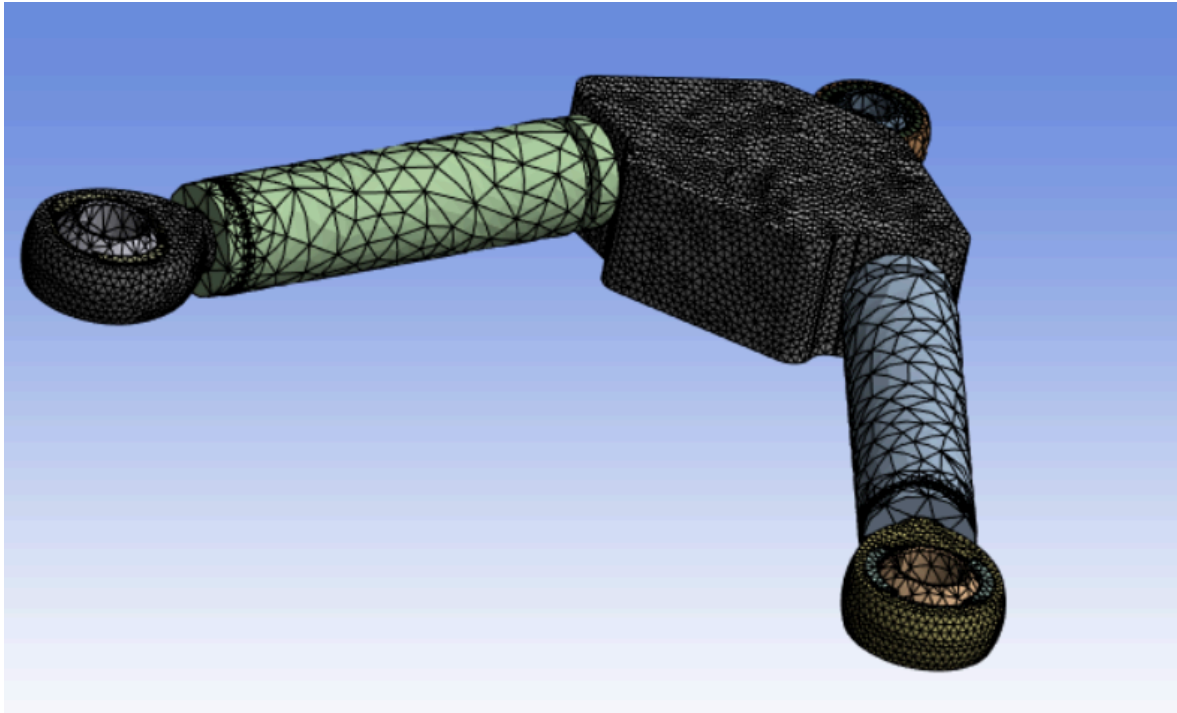


Figure 27: Optimisation Mesh for the Wishbone Clevis

Post-Optimisation Cleanup and Mass Reduction Results

The resulting raw topology generated highly organic load paths. These geometries were exported to SpaceClaim and Fusion 360, smoothed to eliminate non-manufacturable stress risers, and manually redesigned to clean up profiles and ensure clearances for the lower spring-damper bracket.

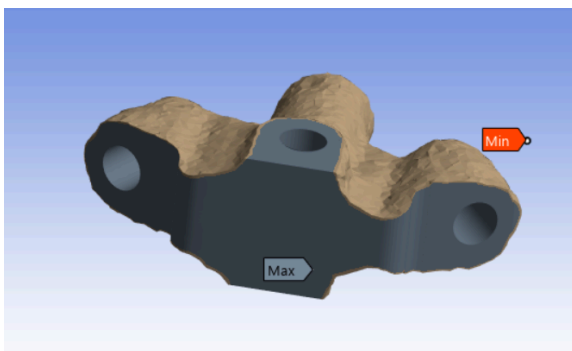


Figure 28: Initial Optimisation model

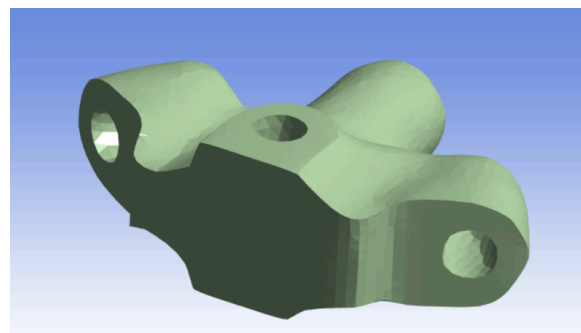


Figure 29: Smoothed out Model

Final stress test

Following geometry reconstruction, the finalized components were subjected to validation testing. Peak stresses rose slightly due to material removal but remained within safe engineering limits for high-tensile alloy steels.

	Stress (Mpa)	Displacement (mm)
Scenario 1	150.7	0.22699
Scenario 2	898.18	1.3528
Scenario 3	551.75	0.83102

The design was then fully smoothed out in fusion360. The upper wishbone clevis was then altered using the same principles

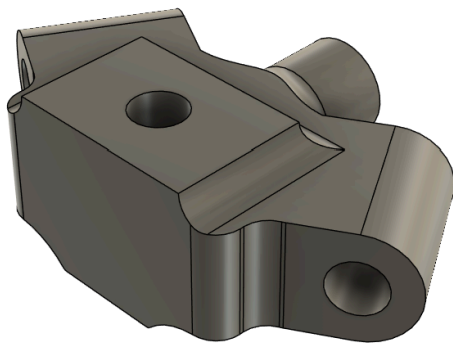


Figure 30: Optimised Lower Wishbone Clevis

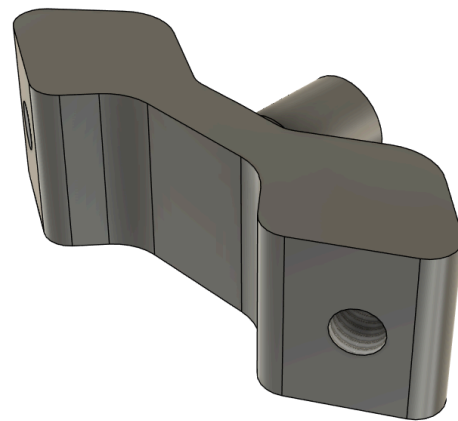


Figure 31: Optimised Upper Wishbone Clevis

	Initial Mass (g)	Optimised Mass (g)	Mass Reduction (%)
Upper Wishbone Clevis	783.553	460.779	41.19%
Lower Wishbone Clevis	608.843	407.052	33.14%
Upper Wishbone	1428.382	1358.839	4.87%
Lower Wishbone	1182.552	859.748	27.30%

Engineering Limitations & Industrial Context

While the static FEA and topology optimisation routines demonstrate successful mass reduction on paper, real-world industrial implementation requires a cautious approach:

Fatigue Strength vs. Static FoS: Automotive suspension linkages are subjected to millions of cyclic load profiles over their operational lifespan. A purely static optimization loop does not account for cyclic fatigue limits or endurance strength.

Stress Concentrations: The organic, varying cross-sections generated via topology optimization naturally present minor surface irregularities. In industrial racing or commercial production, these transitions can act as localized fatigue crack initiation points under continuous track vibrations, demanding substantially higher safety factors than those applied to uniform, tubular structures.

5.6. FEA Conclusions

Following the successful import of the CAD assembly into ANSYS Mechanical, FEA was conducted under the project's defined design load cases to evaluate stress distributions and structural deflections. Through iterative geometry optimisation, material was strategically minimized in low stress regions of the wishbones while ensuring critical areas remained well within allowable material limits. The resulting optimised assembly successfully satisfies the target factor of safety, structurally validating the suspension design for physical fabrication.

6. Conclusions

6.1. General conclusions

This personal project successfully executed a complete engineering design cycle, advancing from a theoretical mathematical framework to a structurally validated, lightweight suspension component.

Dynamic Optimisation: A 2-DOF state-space quarter car model was developed and simulated in Python to evaluate the system's transient response to an 80 mm road disturbance at 10 m/s. The initial target of keeping peak body acceleration under 0.3g while settling within 1 second was mathematically proven to be impossible due to the inherent trade-offs of a passive suspension system. A balanced damping ratio of 0.636 was selected as the optimal compromise, yielding a settling time of 0.999 seconds and a peak combined shock force of 4747.3 N.

Kinematic Modeling: The mathematical parameters were successfully translated into a physical double-wishbone CAD geometry in Fusion 360. To ensure proper steering clearance for the 90 mm track tire under full lock, a 30-degree leg angle was integrated into the wishbone design. Active collision detection validated clean, non-binding travel across the entire stroke of the 50-degree inclined spring-damper unit.

Structural Validation & FEA: Initial static structural FEA in ANSYS Mechanical revealed critical failure zones (up to 2187.8 MPa) at the rod-end threads due to cantilever bending moments and cross-sectional step-downs. Upscaling the chassis connections from M10 to M14 spherical rod ends successfully dropped the peak Von-Mises stresses to a safe 495.26 MPa under the worst-case balanced scenario.

Topology Optimisation: By executing a localized topology optimization routine on the robust M14 design space, parasitic mass was stripped from low-stress regions. After manual geometry reconstruction to ensure manufacturing feasibility, mass was reduced by 41.2% on the upper wishbone clevis and 27.3% on the lower wishbone, structurally validating a high-performance, lightweight assembly.

6.2. Issues with assumptions & Future Work

While the project successfully met its core objectives, several idealized assumptions were made to keep the computational models manageable. Addressing these discrepancies highlights critical pathways for future engineering development:

I. Linear Component Assumptions

- **The Assumption:** The main suspension spring, tire stiffness, and tire damping were modeled as linear constants.
- **The Reality:** Real tires and progressive air/oil dampers (like the Ohlins FSAE or MTB units referenced) exhibit highly non-linear characteristics. At high displacements or near the suspension travel limits, spring rates spike exponentially.
- **Future Work:** The Python state-space framework should be upgraded to a non-linear ODE solver (such as `scipy.integrate.solve_ivp`) incorporating dynamic lookup tables for tire and damper rates to capture high-velocity impact behavior.

II. Continuous Point-Contact & Ground Behavior

- **The Assumption:** The tire acts as a single-point contact spring that never leaves the road surface.
- **The Reality:** While reasonable for a flat Eco-marathon track, severe bumps can induce "wheel hop" where the normal force drops to zero and the tire loses contact with the ground.
- **Future Work:** Introduce a conditional constraint in the simulation code where if the vertical displacement of the unsprung mass exceeds the static tire deflection ($x_u - x_g > \text{static deflection}$), the ground force drops to zero, allowing the model to simulate and mitigate wheel lift.

III. Static FEA vs. Dynamic Fatigue

- **The Assumption:** Structural integrity was validated using static peak loading conditions derived from the maximum transient force step.
- **The Reality:** Suspension wishbones do not fail from single static loads under normal operation; they fail due to fatigue from millions of cyclic track vibrations. The organic, variable cross-sections generated during the topology optimization phase naturally introduce minor surface transitions that can act as dangerous stress-concentration sites under cyclic fatigue.
- **Future Work:** Run a fatigue life sensitivity analysis in ANSYS using a high-cycle load spectrum (S-N curves for high-tensile alloy steel) to guarantee an endurance limit of over 10^6 cycles, ensuring the optimized organic geometries are truly track-safe.

IV. Quarter Car vs. Multi-Body Dynamics

- **The Assumption:** A 1D quarter-car model is sufficient to map the suspension requirements.
- **The Reality:** A quarter-car model isolates vertical motion but completely ignores pitch, roll, and lateral transfer of weight during high-speed cornering or braking.
- **Future Work:** Expand the 2-DOF model into a half-car or full 7-DOF vehicle dynamics model to study how anti-dive, anti-squat, and roll-center migration affect the physical wishbone loading conditions.

7. References

- [1] [What Is Car Suspension? \(How It Works Explained\)](#)
- [2] [Car Suspension Explained: How It Works & Why It Matters - CarXplorer](#)
- [3] [Understanding Quarter Car Model Suspension Systems: Basics And Applications | ShunAuto](#)
- [4] [Between simplicity and accuracy: Effect of adding modeling details on quarter vehicle model accuracy | PLOS One](#)
- [5] Wong, J. Y. (2008). *Theory of Ground Vehicles* (4th ed.). John Wiley & Sons.
- [6] [Development and analysis of passive suspension model with half sine wave road bump for different suspension parameters | AIP Conference Proceedings | AIP Publishing](#)
- [7] [Wishbone Suspension Diagram | Car Anatomy](#)
- [8] [Matlab x SEM: Using Matlab For Suspension System Design - Google Slides](#)
- [9] [How Does Vertical Acceleration Work? | HowStuffWorks](#)
- [10] [Settling Time: What is it? \(And How To Calculate it in MATLAB\) | Electrical4U](#)
- [11] [Understanding Strength and Stiffness in Mechanics of Materials | MachineMFG](#)
- [12] [What is damping coefficient?](#)
- [13] [What is damping ratio?](#)
- [14] [\(DOC\) Modelling and system identification of a quarter car suspension system](#)
- [15] SEM03_datasheet - Session #1.docx.pdf
- [16] [McMaster-Carr](#)

8. Appendix

8.1. Main python code

```
import numpy as np
from scipy import signal
import matplotlib.pyplot as plt
#Define variables
ms=77.5
mu=10
ks=11000
ku=150000
cu=49
```

```

#Initialise Time vector
T=np.linspace(0,5,10000,endpoint=True)
x0=np.array([0,0,0,0])
#change U value
U=np.zeros((10000,2))
#change U (input) to simulate a bump
H=0.08
v=10
L=0.8
t_start=0.5
t_end=t_start+L/v
for i, t in enumerate(T):
    if t_start <= t <= t_end:
        xg=H/2*(1-np.cos((2*np.pi*v)/L*(t-t_start)))
        xg_dot=(np.pi*v*H)/L*np.sin((2*np.pi*v)/L*(t-t_start))
        U[i,0]=xg
        U[i,1]=xg_dot
#store values for later
zeta_list = []
settling_times = []
peak_accelerations_g = []
max_spring_forces = []
max_damper_forces = []
#add zeta value and loop for solver
zeta_range=np.linspace(0.05, 1.5, 100)
for zeta in zeta_range:
    cs=2*zeta*np.sqrt(ks*ms)
#Define matrices
A=np.array([
    [0,0,1,0],
    [0,0,0,1],
    [-ks/ms,ks/ms,-cs/ms,cs/ms],
    [ks/mu,-(ks+ku)/mu,cs/mu,-(cs+cu)/mu]
])
B=np.array([
    [0,0],
    [0,0],
    [0,0],
    [ku/mu,cu/mu]
])
C=np.array([
    [1,-1,0,0],
    [0,0,1,0]
])
D=np.array([
    [0,0],
    [0,0]
])
#set up system
sys=signal.StateSpace(A,B,C,D)

```

```

#run system
tout,yout,xout= signal.lsim(sys, U, T, x0, interp=True)
#extract xs,xu,xs_dot,xu_dot
xs=xout[:,0]
xu=xout[:,1]
xs_dot=xout[:,2]
xu_dot=xout[:,3]
# calculate forces
F_spring= ks*(xs-xu)
F_damper=cs*(xs_dot-xu_dot)
#calculate max forces
F_spring_max=np.max(np.abs(F_spring))
F_damper_max=np.max(np.abs(F_damper))
#peak body acceleration
xs_dotdot=(-F_spring - F_damper)/ms
#settling time
threshold=0.001 #from matlab code
outside_threshold=np.where(np.abs(xs) > threshold)[0]
if len(outside_threshold) > 0:
    settling_time=tout[outside_threshold[-1]]
else:
    settling_time=0
#save values into the tracking arrays
zeta_list.append(zeta)
settling_times.append(settling_time)
peak_accelerations_g.append(np.max(np.abs(xs_dotdot))/9.81) #in g's to compare
to datasheet
max_spring_forces.append(F_spring_max)
max_damper_forces.append(F_damper_max)
#convert lists to array
settling_times=np.array(settling_times)
peak_accelerations_g=np.array(peak_accelerations_g)
max_spring_forces = np.array(max_spring_forces)
max_damper_forces = np.array(max_damper_forces)
zeta_list=np.array(zeta_list)
#Multiple scenario responses
#Scenario 1: Check if both criteria can be met simultaneously (<0.3g AND <1.0s)
ideal_indices = np.where((peak_accelerations_g < 0.3) & (settling_times < 1.0))[0]

#Scenario 2: Settling time balanced between 0.9s and 1.0s
balanced_indices = np.where((settling_times >= 0.9) & (settling_times <= 1.0))[0]

if len(balanced_indices) > 0:
    # From the configurations that settle around 1 second, pick the one with the lowest
    acceleration
    idx_balanced =
    balanced_indices[np.argmin(peak_accelerations_g[balanced_indices])]
else:
    # Fallback: just find the one closest to 1.0s overall
    idx_balanced = np.argmin(np.abs(settling_times - 1.0))

```

```

#Scenario 3: Minimum Settling Time
idx_min_settling = np.argmin(settling_times)

#Scenario 4: Minimum Acceleration
idx_min_accel = np.argmin(peak_accelerations_g)

#Print the Results
#Output 1 - Check to see if best case possible
print("OUTPUT 1: IDEAL SPECIFICATION CHECK (Comfort < 0.3g AND Settling < 1s)")
if len(ideal_indices) > 0:
    print(f"Success! Ideal configuration found at Zeta:
{zeta_list[ideal_indices[0]]:.3f}")
else:
    print("RESULT: SIMULTANEOUS TARGETS MATHEMATICALLY IMPOSSIBLE.")
    print("The system cannot physically achieve <0.3g and settle in <1.0s
simultaneously.")

#Output 2: Balanced case (Used for optimisation)
print("OUTPUT 2: Balanced (Settling Time ~1.0s, Optimized Accel)")
print(f"Optimal Zeta:          {zeta_list[idx_balanced]:.3f}")
print(f"Settling Time:          {settling_times[idx_balanced]:.3f} s  <-- [Target
Met: 0.9s - 1.0s]")
print(f"Peak Acceleration:      {peak_accelerations_g[idx_balanced]:.3f} g <--
[Minimized Compromise]")
print(f"Max Spring Force:        {max_spring_forces[idx_balanced]:.3f} N")
print(f"Max Damper Force:        {max_damper_forces[idx_balanced]:.3f} N")

#Output 3: Minimum Settling time
print("OUTPUT 3: Minimum Settling Time")
print(f"Optimal Zeta:          {zeta_list[idx_min_settling]:.3f}")
print(f"Settling Time:          {settling_times[idx_min_settling]:.3f} s")
print(f"Peak Acceleration:      {peak_accelerations_g[idx_min_settling]:.3f} g")
print(f"Max Spring Force:        {max_spring_forces[idx_min_settling]:.3f} N ")
print(f"Max Damper Force:        {max_damper_forces[idx_min_settling]:.3f} N ")

#Output 4: Minimum peak body acceleration
print("OUTPUT 4: Minimum Peak Body Acceleration")
print(f"Optimal Zeta:          {zeta_list[idx_min_accel]:.3f}")
print(f"Settling Time:          {settling_times[idx_min_accel]:.3f} s")
print(f"Peak Acceleration:      {peak_accelerations_g[idx_min_accel]:.3f} g")
print(f"Max Spring Force:        {max_spring_forces[idx_min_accel]:.3f} N ")
print(f"Max Damper Force:        {max_damper_forces[idx_min_accel]:.3f} N ")

#Plotting x values, chassis displacement (xs) and wheel displacment (xu)
cs_balanced = 2 * zeta_list[idx_balanced] * np.sqrt(ks * ms)
A_balanced = np.array([
    [0, 0, 1, 0],
    [0, 0, 0, 1],
    [-ks/ms, ks/ms, -cs_balanced/ms, cs_balanced/ms],

```

```

[ks/mu, -(ks+ku)/mu, cs_balanced/mu, -(cs_balanced+cu)/mu]
])
sys_balanced = signal.StateSpace(A_balanced, B, C, D)
tout, yout, xout = signal.lsim(sys_balanced, U, T, x0, interp=True)
# Plotting x values, chassis displacement (xs) and wheel displacement (xu)
chassis_displacement=xout[:,0]
wheel_displacement=xout[:,1]
suspension_travel=yout[:,0]
chassis_velocity=yout[:,1]
fig, ax = plt.subplots(2,1)
#Graph 1 for x values
ax[0].plot(tout, chassis_displacement, label="Chassis Displacement ($x_s$)",
color="blue", linewidth=2)
ax[0].plot(tout, wheel_displacement, label="Wheel Displacement ($x_u$)",
color="orange", linestyle="--")
ax[0].set_xlabel("Time (Seconds)")
ax[0].set_ylabel("Displacement (Metres)")
ax[0].legend()
ax[0].set_title("Balanced Dynamic Response")
ax[0].grid(True)
#graph 2.1
line1=ax[1].plot(tout, suspension_travel, label="Suspension Travel ($x_s$-$x_u$)",
color="purple", linewidth=2)
ax[1].set_xlabel("Time (Seconds)")
ax[1].set_ylabel("Displacement (Metres)")
#graph 2.2
ax_twin=ax[1].twinx()
line2= ax_twin.plot(tout, chassis_velocity, label=r"Chassis Velocity
($\dot{x}_s$)", color="green", linestyle="--") #if look weird add r"chassis
ax_twin.set_ylabel("Chassis Velocity (Metres per Second)")
#combine lines
lines=line1+line2
labels=[l.get_label() for l in lines]
ax[1].legend(lines, labels)
ax[1].grid(True)

#plots for zeta against settling time and acceleration
fig, ax1 = plt.subplots()

ax1.plot(zeta_list, peak_accelerations_g, color="red")
ax1.set_xlabel("Damping ratio")
ax1.set_ylabel("Peak acceleration (g)")
ax1.set_title("Peak Body Acceleration and Settling Time vs. Damping Ratio")
ax1.grid(True)

ax2 = ax1.twinx()
ax2.plot(zeta_list, settling_times, color="blue")
ax2.set_ylabel("Settling time (s)")

plt.show()

```

8.2. MATLAB converted code

```
%Define variables
ms=77.5;
mu=10;
ks=11000;
ku=150000;
cu=49;
%Initialise Time vector
T=linspace(0,5,10000);
x0=[0;0;0;0];
%change U value
U=zeros(10000,2);
%change U (input) to simulate a bump
H=0.08;
v=10;
L=0.8;
t_start=0.5;
t_end=t_start+L/v;
for i=1:length(T);
    t=T(i);
    if t >= t_start && t <= t_end
        xg=H/2*(1-cos((2*pi*v)/L*(t-t_start)));
        xg_dot=(pi*v*H)/L*sin((2*pi*v)/L*(t-t_start));
        U(i,1)=xg;
        U(i,2)=xg_dot;
    end
end
%store values for later
zeta_range = linspace(0.05, 1.5, 100);
zeta_list = zeros(1, 100);
settling_times = zeros(1, 100);
peak_accelerations_g = zeros(1, 100);
max_spring_forces = zeros(1, 100);
max_damper_forces = zeros(1, 100);
%add zeta value and loop for solver
for idx = 1:length(zeta_range)
    zeta = zeta_range(idx);
    cs = 2 * zeta * sqrt(ks * ms);
    %Define matrices
    A=[
        0,0,1,0;
        0,0,0,1;
        -ks/ms,ks/ms,-cs/ms,cs/ms;
        ks/mu,-(ks+ku)/mu,cs/mu,-(cs+cu)/mu];
    B= [0,0;
        0,0;
        0,0;
        ku/mu,cu/mu];
    C=[
        1,-1,0,0;
        0,0,1,0];
    D=[0,0;
        0,0];
```

```

%set up system
sys=ss(A,B,C,D);
%run system
[yout,tout,xout]=lsim(sys, U, T, x0);
%extract xs,xu,xs_dot,xu_dot
xs=xout(:,1);
xu=xout(:,2);
xs_dot=xout(:,3);
xu_dot=xout(:,4);
% calculate forces
F_spring= ks*(xs-xu);
F_damper=cs*(xs_dot-xu_dot);
%calculate max forces
F_spring_max=max(abs(F_spring));
F_damper_max=max(abs(F_damper));
%peak body acceleration
xs_dotdot=(-F_spring - F_damper)/ms;
%settling time
threshold=0.001; %from matlab code
outside_threshold=find(abs(xs) > threshold);
if ~isempty(outside_threshold);
    settling_time=tout(outside_threshold(end));
else
    settling_time=0;
end
%save values into the tracking arrays
zeta_list(idx) = zeta;
settling_times(idx) = settling_time;
peak_accelerations_g(idx) = max(abs(xs_dotdot)) / 9.81;
max_spring_forces(idx) = F_spring_max;
max_damper_forces(idx) = F_damper_max;
end
%Multiple scenario responses
%Scenario 1: Check if both criteria can be met simultaneously (<0.3g AND <1.0s)
ideal_indices = find((peak_accelerations_g < 0.3) & (settling_times < 1.0));

%Scenario 2: Settling time balanced between 0.9s and 1.0s
balanced_indices = find((settling_times >= 0.9) & (settling_times <= 1.0));

if ~isempty(balanced_indices)
    [~, local_min_idx] = min(peak_accelerations_g(balanced_indices));
    idx_balanced = balanced_indices(local_min_idx);
else
    [~, idx_balanced] = min(abs(settling_times - 1.0));
end

%Scenario 3: Minimum Settling Time
[~,idx_min_settling] = min(settling_times);

%Scenario 4: Minimum Acceleration
[~,idx_min_accel] = min(peak_accelerations_g);

%print the results
fprintf('OUTPUT 1: IDEAL SPECIFICATION CHECK (Comfort < 0.3g AND Settling < 1s)\n');
if ~isempty(ideal_indices)
    fprintf('Success! Ideal configuration found at Zeta: %.3f\n',
        zeta_list(ideal_indices(1)));
else

```



```

fprintf('RESULT: SIMULTANEOUS TARGETS MATHEMATICALLY IMPOSSIBLE.\n');
fprintf('The system cannot physically achieve <0.3g and settle in <1.0s
simultaneously.\n');
end

%Balanced case (Used for optimisation)
fprintf('OUTPUT 2: Balanced (Settling Time ~1.0s, Optimized Accel)\n');
fprintf('Optimal Zeta: %.3f\n', zeta_list(idx_balanced));
fprintf('Settling Time: %.3f s <-- [Target Met: 0.9s - 1.0s]\n',
settling_times(idx_balanced));
fprintf('Peak Acceleration: %.3f g <-- [Minimized Compromise]\n',
peak_accelerations_g(idx_balanced));
fprintf('Max Spring Force: %.3f N\n', max_spring_forces(idx_balanced));
fprintf('Max Damper Force: %.3f N\n', max_damper_forces(idx_balanced));

%Output 3: Minimum Settling time
fprintf('OUTPUT 3: Minimum Settling Time\n');
fprintf('Optimal Zeta: %.3f\n', zeta_list(idx_min_settling));
fprintf('Settling Time: %.3f s\n', settling_times(idx_min_settling));
fprintf('Peak Acceleration: %.3f g\n', peak_accelerations_g(idx_min_settling));
fprintf('Max Spring Force: %.3f N\n', max_spring_forces(idx_min_settling));
fprintf('Max Damper Force: %.3f N\n', max_damper_forces(idx_min_settling));

%Output 4: Minimum peak body acceleration
fprintf('OUTPUT 4: Minimum Peak Body Acceleration\n');
fprintf('Optimal Zeta: %.3f\n', zeta_list(idx_min_accel));
fprintf('Settling Time: %.3f s\n', settling_times(idx_min_accel));
fprintf('Peak Acceleration: %.3f g\n', peak_accelerations_g(idx_min_accel));
fprintf('Max Spring Force: %.3f N\n', max_spring_forces(idx_min_accel));
fprintf('Max Damper Force: %.3f N\n', max_damper_forces(idx_min_accel));

%Plotting x values, chassis displacement (xs) and wheel displacement (xu)
cs_balanced = 2 * zeta_list(idx_balanced) * sqrt(ks * ms);
A_balanced = [0, 0, 1, 0;
0, 0, 0, 1;
-ks/ms, ks/ms, -cs_balanced/ms, cs_balanced/ms;
ks/mu, -(ks+ku)/mu, cs_balanced/mu, -(cs_balanced+cu)/mu];

sys_sweet = ss(A_balanced, B, C, D);
[yout, tout, xout] = lsim(sys_sweet, U, T, x0);
% Plotting x values, chassis displacement (xs) and wheel displacement (xu)
% Subplot Figure 1: Time Responses
chassis_displacement = xout(:,1);
wheel_displacement = xout(:,2);
suspension_travel = yout(:,1);
chassis_velocity = yout(:,2);
figure('Name', 'Sweet Spot Dynamic Response', 'NumberTitle', 'off');
subplot(2, 1, 1);
plot(tout, chassis_displacement, 'b-', 'LineWidth', 2); hold on;
plot(tout, wheel_displacement, 'r--', 'LineWidth', 1.5);
xlabel('Time (Seconds)');
ylabel('Displacement (Metres)');
title(sprintf('Balanced Dynamic Response (\\zeta = %.3f)',
zeta_list(idx_balanced)));
legend('Chassis Displacement (x_s)', 'Wheel Displacement (x_u)');
grid on;

subplot(2, 1, 2);
yyaxis left

```

```

plot(tout, suspension_travel, 'm-', 'LineWidth', 2);
ylabel('Suspension Travel (Metres)');
xlabel('Time (Seconds)');
yyaxis right
plot(tout, chassis_velocity, 'g--', 'LineWidth', 1.5);
ylabel('Chassis Velocity (m/s)');
legend('Suspension Travel (x_s - x_u)', 'Chassis Velocity (dx_s)');
grid on;

% Figure 2: Damping Sweeping Trade-off Curves
figure('Name', 'Damping Sweep Performance Trade-off', 'NumberTitle', 'off');
yyaxis left
plot(zeta_list, peak_accelerations_g, 'r-', 'LineWidth', 2);
ylabel('Peak Acceleration (g)');
xlabel('Damping Ratio (\zeta)');
grid on;

yyaxis right
plot(zeta_list, settling_times, 'b-', 'LineWidth', 2);
ylabel('Settling Time (Seconds)');
title('Performance Criteria vs. Damping Ratio');

```